Google Cloud H2S

AI Builder Cup

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Team Details

a. Team name: **RetailIQ**
b. Team leader name: **Abhishek Agarwal**
c. Problem Statement: Retail & Commerce: Intelligent Customer and Business Experiences

Team member: Vivek Sharma

Brief about the idea

RetaiIQ: shelf evidence to an accountable retail action

A CPG field representative captures the shelf, verifies a stock discrepancy, prepares a source-linked buyer conversation and saves an action a manager can review. The buyer retains commercial authority.

SENSING

Qualified shelf observations; PASS, FAIL or NOT VISIBLE.

EXECUTION

Locate/reconcile existing stock or justify replenishment.

EMPOWERMENT

Cited facts, rehearsal, structured debrief and a reviewable record.

Primary user: CPG field rep. Managers review; buyers approve; shoppers discover.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

1. How it differs

Connects shelf evidence, stock logic, a prepared conversation and manager review in one visit. It goes beyond an isolated audit score or chat answer.

Design comparison; no exhaustive competitor benchmark is claimed.

2. How it solves the problem

Zero facings plus 64 recorded units triggers a locate-and-reconcile task. A justified order nets usable and inbound stock against demand, case rules and shelf life.

The example is synthetic; physical verification remains necessary.

3. USP

Evidence + retail arithmetic + human control. Own-brand compliance and competitor benchmarking stay separate. Unseen evidence remains unknown; a saved record exposes the proposed action.

Research rationale: a 55-retailer study links availability to sensing, execution and empowerment. Corsten et al., IJRDM, May 2026. Industry findings are not RetailIQ outcomes.

List of features offered by the solution

Field representative

Shelf photo/video analysis; own-brand audit and separate competitor benchmark.

Guided counts, stock discrepancy checks and forecast-aware proposed orders.

Source-linked pitch, buyer rehearsal and structured debrief.

Manager and buyer

Saved audits and proposed orders with originating visit context.

Revenue, forecasts, inventory alerts and segment reference views.

Optional private evidence in backend mode; human commercial approval.

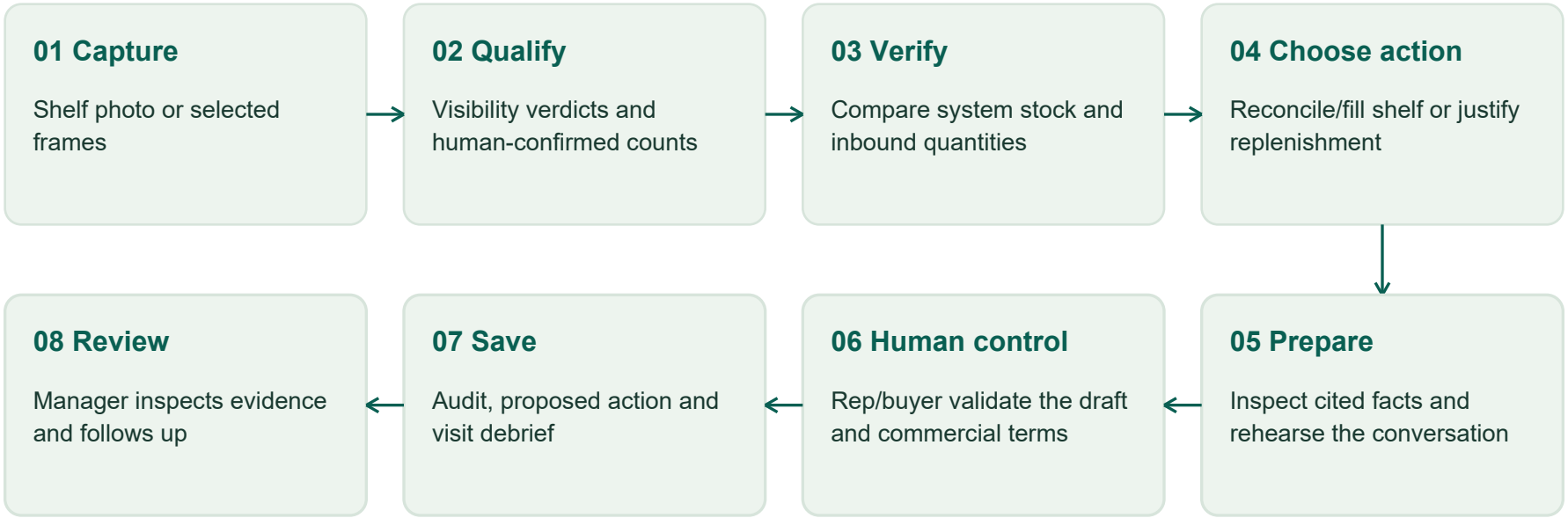
Shopper

Conversational catalogue discovery by need, preference and basket context.

Compare product stock, price and nutrition.

Synthetic figures are labelled; recorded mode requires no model key.

Process flow diagram or Use-case diagram



Decision branch: empty shelf + positive recorded stock → locate and reconcile first. Verified replenishment need → demand minus usable stock and inbound, bounded by case size and shelf life.

A proposed order remains a draft; no automatic ERP placement is claimed.

Wireframes/Mock diagrams of the proposed solution (optional)

Field Execution

- Store / buyer / visit context
- Shelf evidence + verdicts
- Counts + stock constraints
- **Review next action; save visit**

Team and Analytics

- Revenue + inventory risk
- Forecast + segment context
- Saved audit / order events
- **Inspect evidence; follow up**

Shopper Insights

- Need + preferences + basket
- Conversation + options
- Stock + price + nutrition
- **Compare suitable products**

Explanatory screen structures; actual prototype captures appear in the snapshot section.

Architecture diagram of the proposed solution

CURRENT PUBLIC PROFILE

Firebase Spark

Unbilled hosting; billing disabled.

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RetailIQ browser

Recorded mode; local event storage.

↓

Optional AI Studio

Confirmed unbilled key; quota stop.

IMPLEMENTED BACKEND REFERENCE

Cloud Run / FastAPI

Authenticated API; bounded requests and validated contracts.

Gemini on Vertex AI

Backend text + shelf vision

BigQuery / BQML

Analytics, retrieval, forecasts

Firestore

Visit events + review metadata

Cloud Storage

Optional private shelf evidence

Reference backend was historically verified; its retired endpoint is not the current static host.

Model credentials stay off the browser in backend mode. Private IAM is the backend default.

Technologies to be used in the solution

Technology	Purpose	Actual status
Firebase Hosting (Spark)	Public static app and video	Current host; project billing disabled
HTML / CSS / JavaScript	Rep, manager and shopper UI	Current browser application
Gemini / AI Studio	Optional direct text and vision	Unbilled-key confirmation; quota stop
Cloud Run / FastAPI	Authenticated application API	Implemented backend reference
Gemini on Vertex AI	Backend model inference	Historical 3.6 Flash reference evaluated
BigQuery / BigQuery ML	Analytics, retrieval, ARIMA forecasts	Synthetic history and holdout checked
Firestore	Saved visit events and review metadata	Implemented backend persistence
Cloud Storage / Secret Manager	Private evidence / credentials	Implemented backend safeguards
Playwright / pytest / CI	Software and evaluator regressions	Local checks and workflow supplied

No usage of Vertex AI Search, Looker, AlloyDB or Cloud SQL is claimed.

Estimated implementation cost (optional)

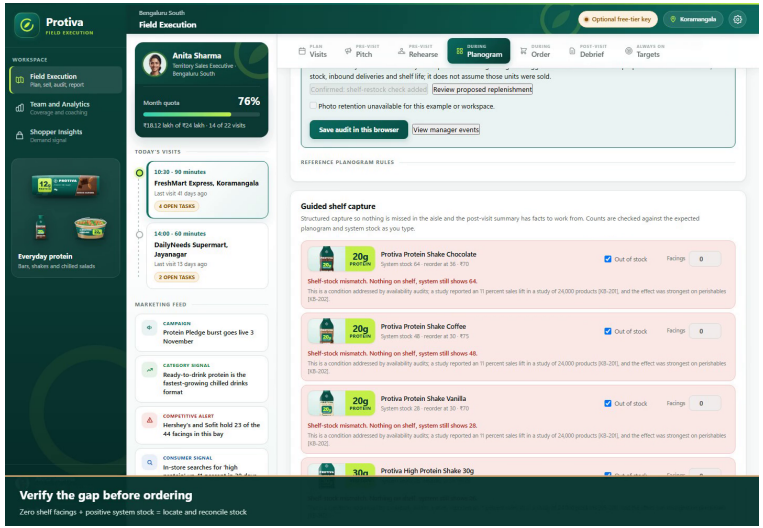
Current prototype: Firebase Spark project billing is disabled; recorded mode makes no AI calls. Optional company services are metered and consume credits.

Editable planning assumption	Scenario value
10 reps × 6 visits/day × 22 days	1,320 visits / month
Assumed 12-minute baseline → 5-minute scoped-task target	7 minutes / visit
1,320 × 7 / 60	154 hours released capacity
154 hours × assumed INR 300 / hour	INR 46,200 capacity value
Target operating budget: INR 5 × 1,320 visits	INR 6,600 / month
Illustrative capacity value less operating target	INR 39,600 / month

Implementation estimate: integration, training and device scope require discovery; no engineering cost is invented. The scenario is not measured ROI or guaranteed cash saving. The operating budget is a target, not an invoice.

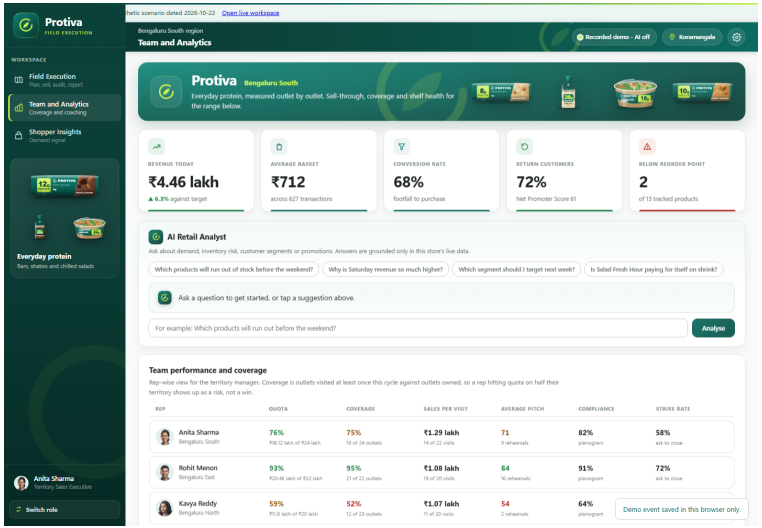
Scope: audit preparation and debrief administration, not the entire store visit.

Snapshots of the prototype



Field rep: confirm the shelf-stock mismatch

Actual prototype; synthetic zero facings versus 64 recorded units. Human stock verification precedes a new purchase.



Manager: review context and saved events

Actual prototype; synthetic KPIs and team figures. Current free-host events remain in the browser.

Prototype Performance report/Benchmarking

Area	Result / baseline	What the result establishes
Current public host	18 HTTP/browser checks passed	Hashes, private paths, mobile fit; no AI calls
Public video player	6 checks; native browser playback	222.44-second playable MP4; no AI/backend calls
Browser / policy	157 + 7 checks passed	Exercised UI contracts; mocked backend responses
Python regressions	60 checks passed	51 API + 4 deploy doubles + 5 free-host guards
Offline evaluator	38 deterministic assertions	Facts and corrupt-response rejection; no model calls
Historical live model	10/10 synthetic cases	Vertex Gemini 3.6 Flash; not current free-model proof
Historical cloud	22 deployed checks passed	Warehouse, events, private photo; endpoint retired
Synthetic forecast	WAPE 1.249% vs baseline 2.824%	7-day generated holdout; not real-store accuracy

Credential-free reports: submission/evidence/. No achieved sales uplift, production forecast accuracy, visual-counting precision or measured operating cost is claimed.

Additional Details/Future Development (if any)

Proposed development

1. Replace generated data with permissioned POS, stock and inbound feeds.

2. Independently label shelf evidence and measure precision, coverage and abstention.

3. Add approved enterprise identity and ERP draft-order handoffs.

4. Evaluate the chosen current model and establish cost/latency under representative usage.

Falsifiable pilot

Compare a two-week baseline with an assisted period.

Test the five-minute target for scoped audit/debrief work. Measure corrections and productive reuse of time.

Track stock discrepancy closure, source-supported claims, order waste risk, latency and cost per completed workflow.

Roadmap and pilot targets only. No achieved revenue uplift or stockout reduction is claimed.

Provide links to your:

- GitHub Public Repository https://github.com/abhishek-0706/AI-BUILDER-CUP-2026
- Demo Video Link (3 Minutes) https://retailiq-free-20261018-7ce297.web.app/demo.html
- Final Product Link https://retailiq-free-20261018-7ce297.web.app/?demo=recorded

Submission access and scope

Public source, product and Google-hosted video links are verified without authentication.

Video: 3:42.44, meeting the online form's 3–4-minute range; AI-assisted narration from the creator-provided voice sample. The walkthrough is recorded with synthetic commercial context.

Current public product: Firebase Spark, project billing disabled, browser-local saves. A no-key recorded visit requires no model request.

Team RetailIQ — Abhishek Agarwal (team leader); Vivek Sharma (team member).